SOP 1

Start-up and shut-down

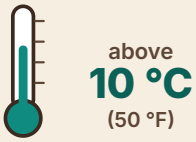
When: Every day you cut **Time:** 10 min to start, 10 min to close (Estimate)

Who: Whoever runs the laser that day

You need: Room thermometer | Lens tissue and lens cleaner | Brush and bin | Planner tablet

Start of the day

1

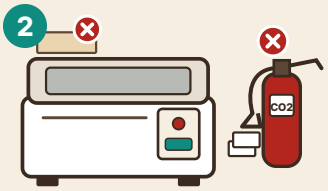


Check the room temperature.

The water loop must stay above 10 °C (50 °F). Cold, damp wood also fits differently.

- Below 10 °C: do not start. Warm the room first.

2

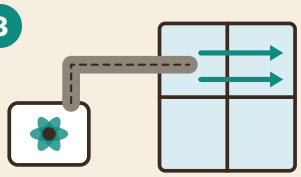


Clear around the laser.

Nothing on the lid. No paper, offcuts or rags next to the machine.

- CO2 extinguisher and fire blanket within reach.

3

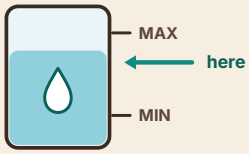


Open the vent. Switch on the exhaust.

Run it before, during and after every job.

- A tissue held at the intake gets pulled in.

4

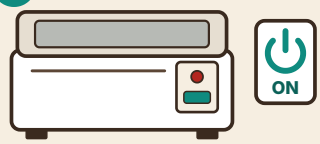


Look at the cooling water.

Clear, between MIN and MAX, no bubbles in the tube.

- Low or cloudy: top up with distilled water (SOP-5).

5

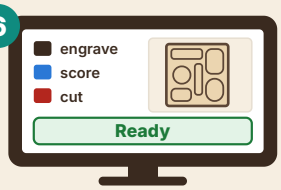


Turn on the laser.

Wait until the head moves home and stops.

- No error on the control panel.

6

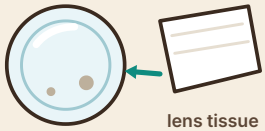


Open LightBurn and connect.

Use the saved device profile. Do not change layer settings by hand.

- Status shows Ready.

7



Look at the lens.

A smoke film or dots on the lens make weak cuts. Wipe with lens tissue and cleaner.

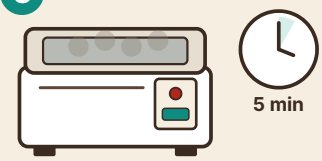
- Clean, no cracks, no pits.

Stop and fix first if

- Smoke smell in the room while a job runs: the exhaust is not pulling.
- The water is warm, low, or has bubbles in the tube.
- The control panel shows an error, or the head does not home.

End of the day

8

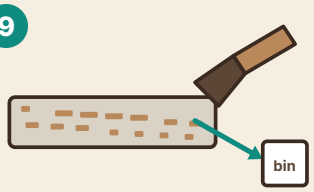


Let the smoke clear.

Leave the exhaust on for 5 minutes after the last job.

- No smoke smell when you lift the lid.

9

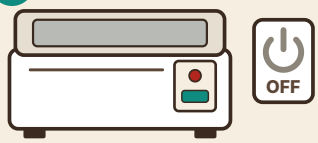


Empty the crumb tray. Sweep the bed.

Small scraps are fuel for a fire.

- Tray empty, bed clear, bin emptied outside.

10



Switch off: laser first, then exhaust.

Close the window vent.

- Winter: the room stays above 10 °C overnight, or drain the water loop.

11



Log the day in the planner.

Mark the batches you cut. Log any problem on the Quality tab.

- Every batch cut today shows Cut.

Hartwood Designs, SOP-1, Rev A, 2026-10-12. Times and settings marked Estimate are planning numbers until measured on the real machine. Machine details follow the OMTech manual when it differs from this sheet.